

Lachlan Biersteker – Engineering Physics

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SUMMARY

Deans list second-year Engineering Physics student at the University of British Columbia with hands-on experience in embedded systems, C/C++ programming, data analysis and machine learning in python, as well as PCB and CAD design. Seeking an engineering internship to apply physics and programming fundamentals to impactful problems.

EDUCATION

University of British Columbia — Vancouver, BC

Sept 2025 – Present

B.A.Sc., Engineering Physics (Second Year) · Expected Graduation 2030

- Relevant coursework: Differential & Integral Calculus, Linear Algebra, Programming in C/C++, Quantum Mechanics, Thermodynamics & SHM, Electromagnetism, AC/DC Circuits.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, MATLAB/Simulink

Embedded & Hardware: STM32, Nordic, Arduino, I2C, SPI, UART

Design & CAD: Altium, SolidWorks, KiCad

Tools: VS Code, PlatformIO, Git/GitHub, STM32CubeIDE, IntelliJ IDEA

Certifications: Altium PCB design · IBM Basics of Quantum Information · MATLAB Onramp – MathWorks

TECHNICAL PROJECTS

Wearable Motion-Sensing System for Machine Learning Exercise Form Analysis · [GitHub](#)

- Effectively built a wearable device that automatically detects reps in an exercise and tells user how they can correct their form for each rep using 2 machine learning models trained on hours of exercise and IMU data.
- Programmed microcontroller in C++, developed signal processing and machine learning in python achieving 85% segmentation accuracy for rep detection and 95% on form rating metrics.
- Fabricated and efficiently performed bring up of custom Altium designed compact PCB to stream data via Bluetooth.

Self Stabilizing Plane & STM32 Flight Motion Sensing System · [GitHub](#)

- Developed embedded firmware in C using STM32CubeIDE and HAL libraries to interface an IMU and barometric pressure sensor over I2C and UART as well as two servos as ailerons controlled using PID tuning logic.
- Acquired, processed, and debugged real-time sensor data over UART and implemented sensor fusion and complimentary filter to give clean values for roll pitch and yaw used in the PID controller.

SolidWorks Prototype Toothpaste Dispenser for Disabled Individuals · [GitHub](#)

- Collaborated with a team of classmates to design a toothpaste dispenser in SolidWorks for individuals with limited motor control for a school project.
- Conducted force analysis on the design to validate the design's robustness.
- Created a technical memo and delivered an oral presentation of the device's features to the class.

Avionic sensor PCB · [GitHub](#)

- Successfully integrated various bare chip avionic sensors into a 4 layer mixed signal PCB in Altium with an STM32 microchip through I2C, applying impedance controlling and an antenna bias network for GPS connection.

EXPERIENCE & LEADERSHIP

Ski Instructor — WinSport, Calgary, AB

Winter 2022/23, 2023/24

- Coached skiing to ages 3–18+ of all skill levels, enforcing safety protocols and adapting to dynamic conditions; CSIA certified.

Bike Instructor — Pedalheads, Calgary, AB

Summer 2023, 2024

- Instructed children ages 5–12 in cycling and roadway safety, prioritizing adaptability and student safety taking hundreds of children from training wheels to biking on their own every summer.

Basketball Coach (Grade 8 Team) — Masters Academy

Dec 2024 – Mar 2025

- Ran practices and coached games, developing players' skills and team strategy; leading the team to a first place finish in the league.